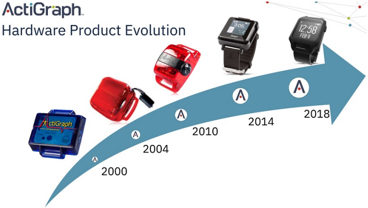
Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 4

Devices

Clinical and Research devices often require wired connections, but wireless is taking over as security is addressed.



Data

Typically time based

- A header often defines parameters
- Each row is an entry
- Each column is a measure

Sometimes the data are stored in a mixed format, text and low level binary

Log Record Types

Note that some undocumented records are used for internal state or testing. They may be safely ignored.

ID (dec)	ID (hex)	Type	Description
0	0x00	ACTIVITY	One second of raw activity samples packed into 12-bit values in YXZ order.
2	0x02	BATTERY	Battery voltage in millivolts as a little-endian unsigned short (2 bytes).
3	0x03	EVENT	Logging records used for internal debugging. These records notably contain information about idle sleep mode. When entering idle sleep mode, a record with payload 0x08 is created. When existing idle sleep mode, a record with payload 0x09 is created.
4	0x04	HEART_RATE_BPM	Heart rate average beats per minute (BPM) as one byte unsigned integer.
5	0x05	LUX	Lux value as a little-endian unsigned short (2 bytes).
6	0x06	METADATA	Arbitrary metadata content. The first record in every log is contains subject data in JSON format.
7	0x07	TAG	13 Byte Serial, 1 Byte Tx Power, 1 Byte (signed) RSSI
9	0x09	EPOCH	60-second epoch data
11	0x0B	HEART_RATE_ANT	Heart Rate RR information from ANT+ sensor.
12	0x0C	EPOCH2	60-second epoch data
13	0x0D	CAPSENSE	Capacitive sense data
14	0x0E	HEART_RATE_BLE	Bluetooth heart rate information (BPM and RR). This is a Bluetooth standard format.
15	0x0F	EPOCH3	60-second epoch data

Typical Analysis

- Measures of central tendency
- Measures of variability
- Comparisons
- Trends (Binned, Change Point, Detrended Fluctuation)

Central Tendency

- Arithmetic Mean:
  $\bar{A} = \frac{1}{n}\sum_{i=1}^{n} a_i$
- Geometric Mean[1]:
  $\bar{G} = (\prod_{i=1}^{n} x_i)^{\frac{1}{n}} = \sqrt[n]{x_1 x_2 \ldots x_n}$
- Harmonic Mean[2]:
  $\bar{H} == \frac{n}{\frac{1}{x_1}+\frac{1}{x_2}+\ldots\frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n}\frac{1}{x_i}}$
- Median
- Mode

[1] useful whenever the quantities to be averaged combine multiplicatively

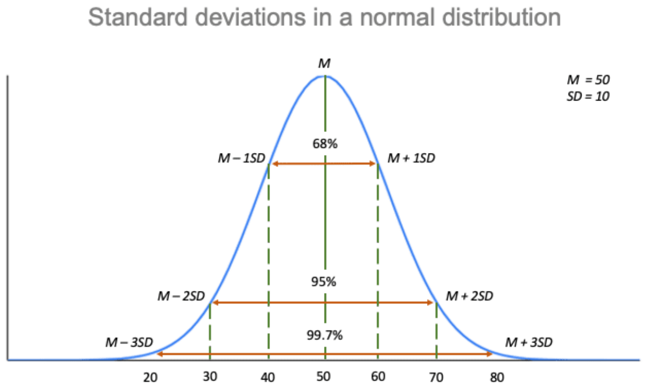
[2] most appropriate average for ratios and rates such as speeds, robust to outliers

Measures of Variance

- Range:
  $\text{range} = max(X) - min(X)$
- Variance:
  $\text{Var}(X) = \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$
- Standard Deviation:
  $\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$
- Pooled Standard Deviation:
  $\sigma_p = \sqrt{\frac{\sum_{i=1}^{k}(n_i - 1)\sigma_i^2}{\sum_{i=1}^{k}(n_i - 1)}} = \sqrt{\frac{(n_1-1)\sigma_1^2 + (n_2-1)\sigma_2^2 + \ldots (n_k-1)\sigma_k^2}{n_1+n_2+\ldots+n_k-k}}$

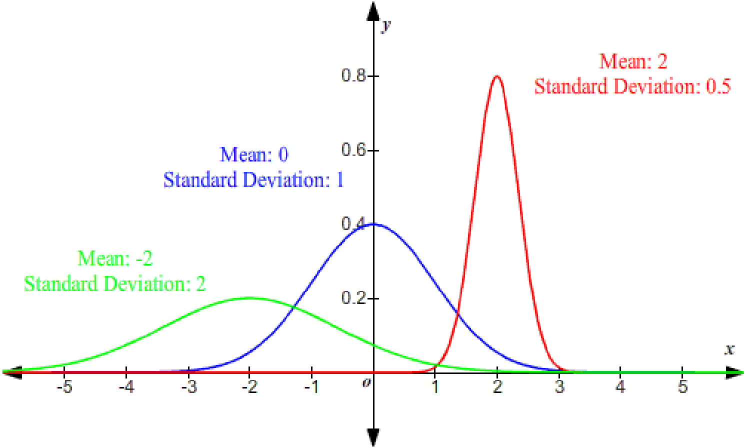
Standard Deviation

This is a "normal" curve, coming from theory driven by observations. One standard deviation is the inflection point on the curve.



Pooled Standard Deviation

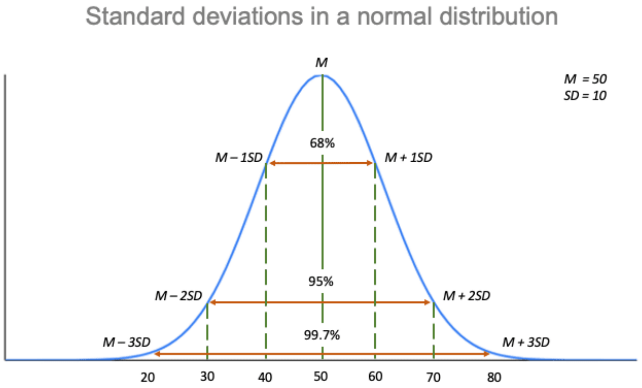
Suppose we have three participants and have measured the number of steps they take every day for a month. We have an average for each, and a standard deviation around those means for each participant. So then, how do we calculate the 'average' mean and standard deviation?



Standardized Measures (Z-Score)

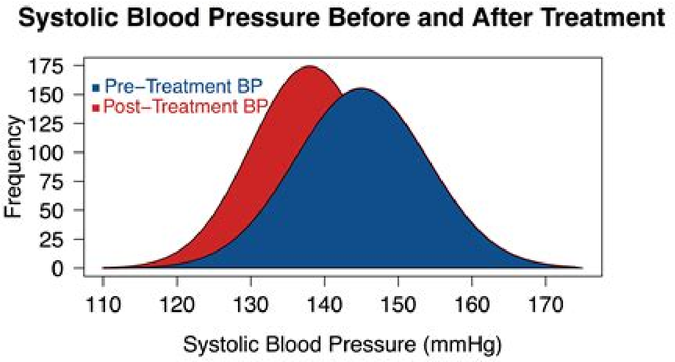
What if we wanted to make some comparisons when measures are in different units? Or what what if the samples seem to have some bias, and we want to understand deviations from that bias?

$$z_i = \frac{x_i - \mu}{\sigma}$$



Comparison Tests

Suppose that we have samples of blood pressure before a treatment, and then after.
Likewise, what if we had two different groups we wanted to compare?



Comparison Tests

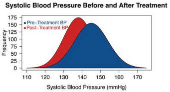
Suppose that we have samples of blood pressure before a treatment, and then after.
Likewise, what if we had two different groups we wanted to compare?
We can use a t-test! This tests if the two samples are likely to have been sampled from the same population.

$$df = n_1 + n_2 - 2$$

$$\sigma_p = \sqrt{\frac{(n_1-1)\sigma_1^2+(n_2-1)\sigma_2^2}{df}}$$

$$t = \frac{(\mu_1-\mu_2)}{\sigma_p \cdot \sqrt{(\frac{1}{n_1})+(\frac{1}{n_2})}}$$

$$cdf = tcdf(t, df)$$



	Upper tail area				
df	0.1	0.05	0.025	0.01	0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Adapted from Table III of R. A. Fisher and F. Yates, *Statistical Tables for Biological, Agricultural and Medical Research*, Sixth Edition, Pearson Education Limited, © 1963 R. A. Fisher and F. Yates.

Analysis of Variance (ANOVA)

And what if we have three or more groups? A t-test isn't really going to work for us... why?
We can test if there exists a statistically significant difference among the groups first.

group	1	2	3
	15	17	9
	12	5	12
	12	15	11
	9	12	11
		20	8
		14	
n	5	7	6
$\sum$	49	85	54

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

H_1 : H_0 is untrue

$\alpha = 0.05$ (1/20 probability what we observe is from chance alone)

Analysis of Variance (ANOVA)

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group	1	2	3
	15	17	9
	12	5	12
	12	15	11
	9	12	11
		20	8
		14	
n	4	6	5
$\sum$	48	83	51

$N = \sum_{i=1}^{k} n_i = 15$

Total Sum of Squares:

$$SS_T = \sum(X - \bar{(}X)^2) = \sum X^2 - \frac{(\sum X)^2}{N}$$

Sum of Squares Between Groups:

$$SS_B = \sum n_i(\bar{X}_i - \bar{X})^2 = \sum_{i=1}^{k} \left(\frac{(\sum X_i)^2}{n_i} \right) - \frac{(\sum X)^2}{N} = \frac{48^2}{4} + \frac{83^2}{6} + \frac{51^2}{5} - \frac{182^2}{15}$$

Sum of Squares Within Groups:

$$SS_W = SS_T - SS_B$$

Analysis of Variance (ANOVA)

And what if we have three or more groups? A t-test isn't really going to work for us... why?
We can test if there exists a statistically significant difference among the groups first.

MS is our mean squares

$$m = \text{number of groups}$$
$$N = \text{number of observations}$$
$$n_j = \text{number of observations in group j}$$
$$x_{ij} = \text{measure for observation i group j}$$
$$\bar{x}_j = \text{mean for jth group}$$
$$\bar{X} = \text{overall mean} = \frac{\sum x_{ij}}{N}$$

$$SS_{total} = \sum^{N} (x_{ij} - \bar{X})^2 = \sum x_{ij}^2 - \frac{(\sum x_{ij})^2}{N}$$

Repeated Measures ANOVA

Suppose we have made measures at several time points on a group of subjects, such that we have something like this:

$$
\begin{aligned}
n &= \text{number of subjects}\\
k &= \text{number of conditions}\\
x_{ij} &= \text{measure for subject i condition j}\\
\bar{x}_j &= \text{mean for ith condition}\\
\bar{x}_i &= \text{mean for jth subject}\\
\bar{X} &= \text{overall mean}
\end{aligned}
$$

$$SS_{subjects} = \sum_{i=1}^{n}\sum_{j=1}^{k}(x_{ij} - \bar{x}_i)^2$$

$$SS_{conditions} = n\sum_{j=1}^{k}(\bar{x}_j - \bar{X})^2$$